

Exercise 2: Error Handling

Scenario 1: Handle exceptions during fund transfers between accounts

Question: Write a stored procedure **SafeTransferFunds** that transfers funds between two accounts. Ensure that if any error occurs (e.g., insufficient funds), an appropriate error message is logged and the transaction is rolled back.

Code:

```
CREATE OR REPLACE PROCEDURE SafeTransferFunds(
    p_FromAccount IN NUMBER,
    p_ToAccount IN NUMBER,
    p_Amount IN NUMBER
)
IS
    v_Balance NUMBER;
BEGIN

    SELECT Balance
    INTO v_Balance
    FROM Accounts
    WHERE AccountID = p_FromAccount;

    IF v_Balance < p_Amount THEN
        RAISE_APPLICATION_ERROR(-20001,
            'Insufficient Funds!');
    END IF;

    UPDATE Accounts
    SET Balance = Balance - p_Amount
    WHERE AccountID = p_FromAccount;

    UPDATE Accounts
    SET Balance = Balance + p_Amount
    WHERE AccountID = p_ToAccount;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Funds Transferred Successfully.');
```

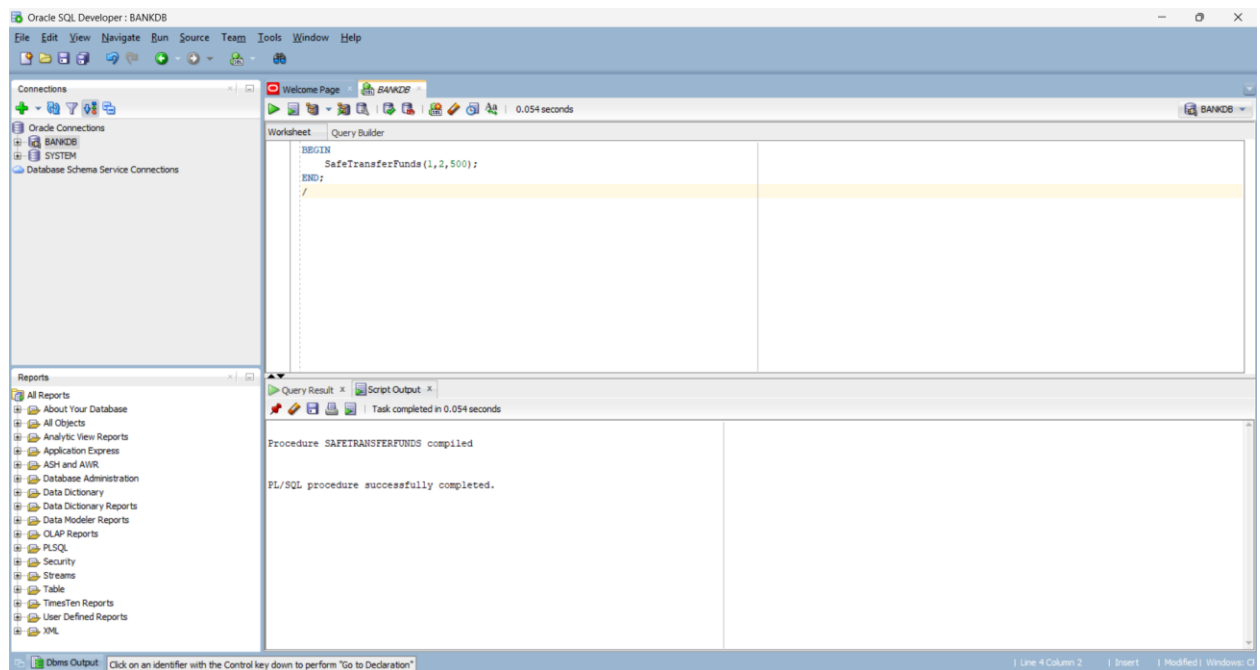
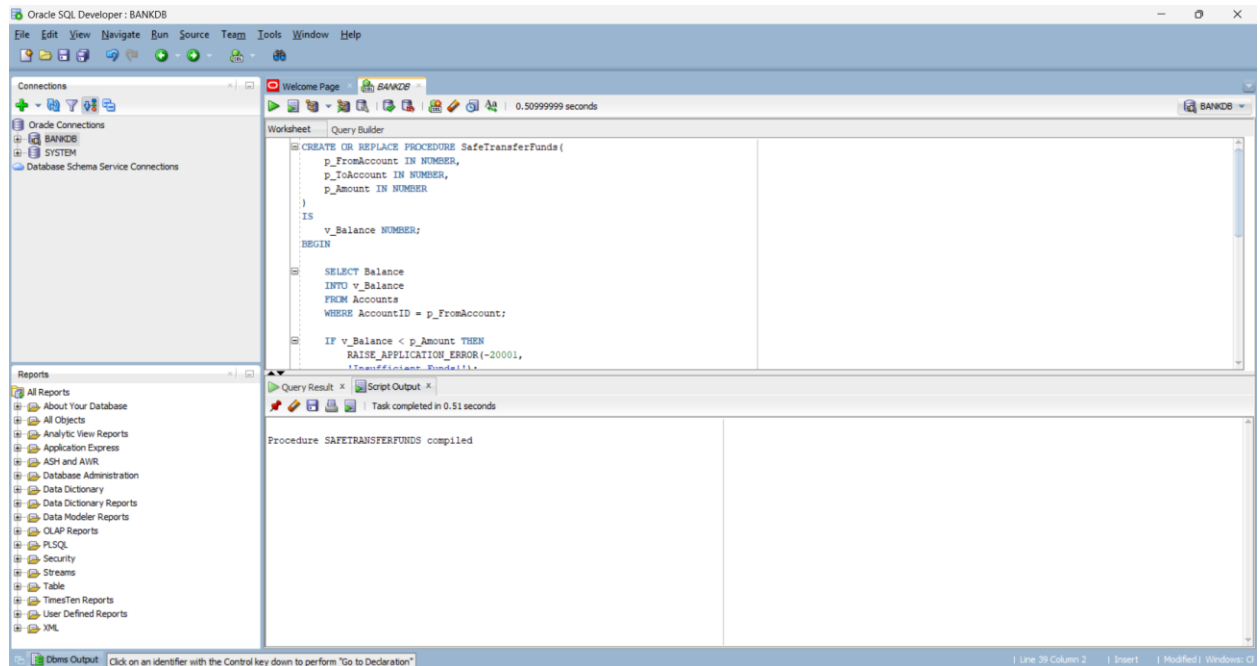
EXCEPTION

```
    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
```

```
END;  
/  

```

Output:



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Connections

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Worksheet

```
SELECT * FROM Accounts;
```

Reports

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Script Output x Query Result x

All Rows Fetched: 2 in 0.01 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE	LASTMODIFIED
1	1	1	Savings	500	10-JUL-26
2	2	2	Checking	2000	10-JUL-26

Dbms Output Click on an identifier with the Control key down to perform "Go to Declaration"

Line 1 Column 24 Insert Modified Windows

Scenario 2: Manage errors when updating employee salaries.

Question: Write a stored procedure **UpdateSalary** that increases the salary of an employee by a given percentage. If the employee ID does not exist, handle the exception and log an error message.

Code:

```
CREATE OR REPLACE PROCEDURE UpdateSalary(

    p_EmployeeID IN NUMBER,
    p_Percentage IN NUMBER

)
IS

    v_Count NUMBER;

BEGIN

    SELECT COUNT(*)
    INTO v_Count
    FROM Employees
    WHERE EmployeeID = p_EmployeeID;

    IF v_Count = 0 THEN

        RAISE_APPLICATION_ERROR(-20002,
            'Employee ID Not Found.');
```

```
    END IF;

    UPDATE Employees
    SET Salary = Salary + (Salary * p_Percentage/100)
    WHERE EmployeeID = p_EmployeeID;

    COMMIT;

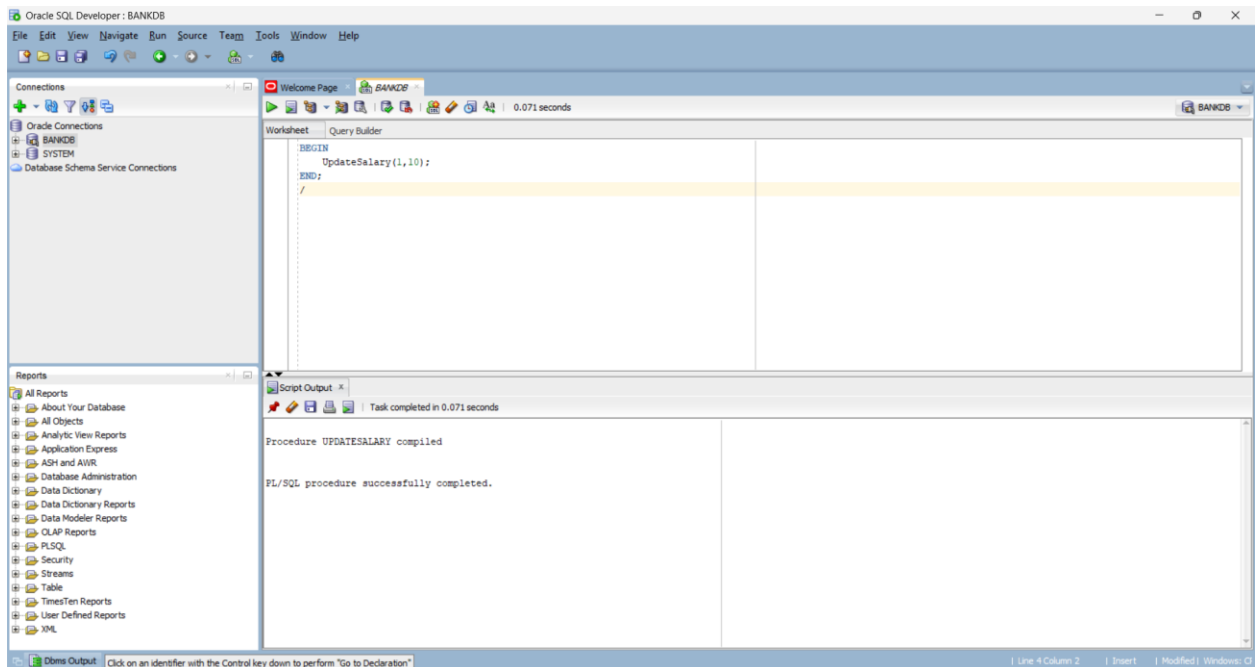
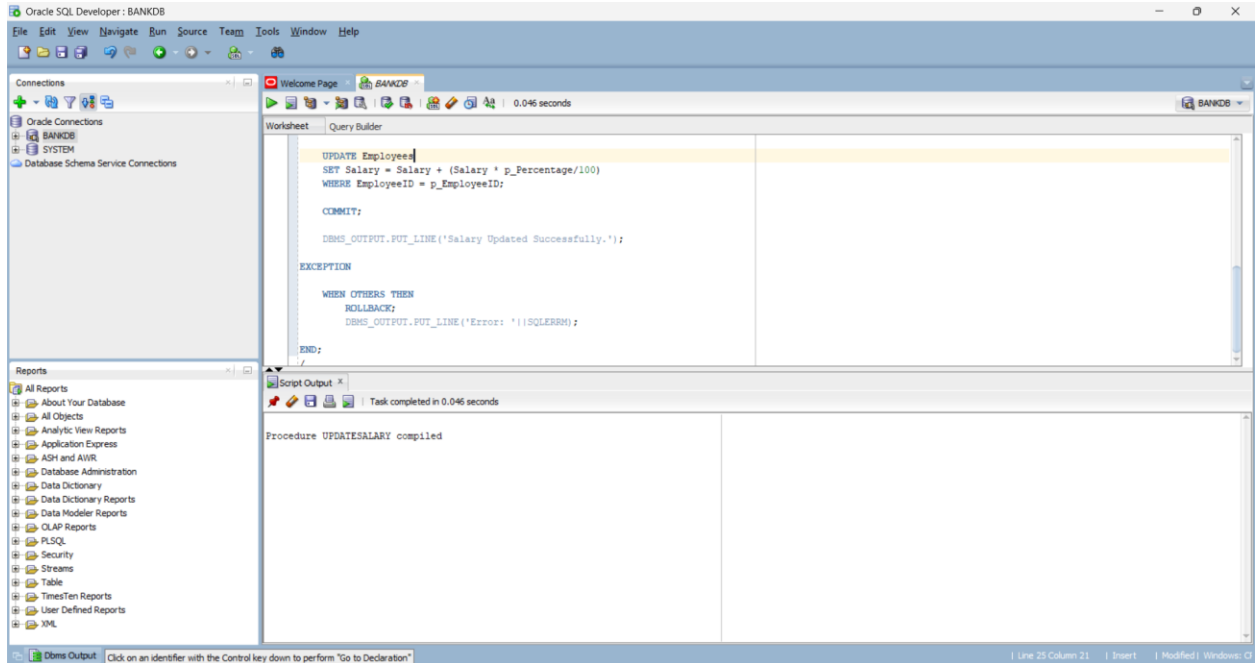
    DBMS_OUTPUT.PUT_LINE('Salary Updated Successfully.');
```

```
EXCEPTION

    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;
```

Output:



Oracle SQL Developer : BANKDB

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Connections

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Worksheet

```
SELECT *  
FROM Employees;
```

Reports

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- XML

Script Output

Query Result

All Rows Fetched: 2 in 0.037 seconds

EMPLOYEEID	NAME	POSITION	SALARY	DEPARTMENT	HIREDATE
1	Alice Johnson	Manager	77000	HR	15-JUN-15
2	Bob Brown	Developer	60000	IT	20-MAR-17

Dbms Output

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 2 Column 16 | Insert | Modified | Windows

Scenario 3: Ensure data integrity when adding a new customer.

Question: Write a stored procedure **AddNewCustomer** that inserts a new customer into the Customers table. If a customer with the same ID already exists, handle the exception by logging an error and preventing the insertion.

Code:

```
CREATE OR REPLACE PROCEDURE AddNewCustomer(  
  
    p_CustomerID IN NUMBER,  
    p_Name IN VARCHAR2,  
    p_DOB IN DATE,  
    p_Balance IN NUMBER  
  
)  
IS  
  
BEGIN  
  
    INSERT INTO Customers(  
  
        CustomerID,  
        Name,  
        DOB,  
        Balance,  
        LastModified  
  
    )  
  
    VALUES(  
  
        p_CustomerID,  
        p_Name,  
        p_DOB,  
        p_Balance,  
        SYSDATE  
  
    );  
  
    COMMIT;  
  
    DBMS_OUTPUT.PUT_LINE('Customer Added Successfully.');
```

EXCEPTION

```
    WHEN DUP_VAL_ON_INDEX THEN
```

```
ROLLBACK;
```

```
DBMS_OUTPUT.PUT_LINE(  
    'Customer ID already exists.');
```

```
WHEN OTHERS THEN
```

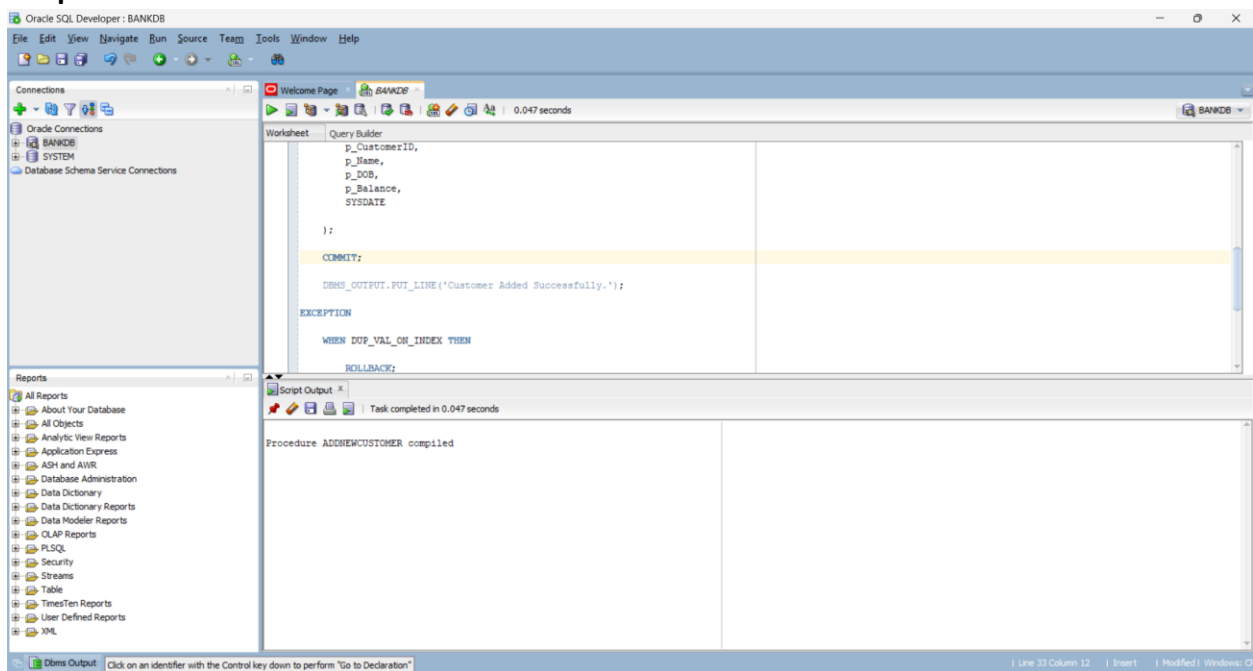
```
ROLLBACK;
```

```
DBMS_OUTPUT.PUT_LINE(  
    'Error: ' || SQLERRM);
```

```
END;
```

```
/
```

Output:



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Worksheet

```
BEGIN
AddNewCustomer(
3,
'David',
TO_DATE('2000-06-10','YYYY-MM-DD'),
25000
);
END;
```

Script Output

Task completed in 0.089 seconds

Procedure ADDNEWCUSTOMER compiled

PL/SQL procedure successfully completed.

Dbms Output Click on an identifier with the Control key down to perform "Go to Declaration"

Line 13 Column 2 Insert Modified Windows

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Worksheet

```
SELECT *
FROM Customers;
```

Script Output

Query Result: 3 in 0.009 seconds

CUSTOMERID	NAME	DOB	BALANCE	LASTMODIFIED	ISVIP
1	3 David	10-JUN-00	25000	20-JUL-26	(null)
2	1 John Doe	15-MAY-50	1000	10-JUL-26	(null)
3	2 Jane Smith	20-JUL-90	1500	10-JUL-26	(null)

Dbms Output Click on an identifier with the Control key down to perform "Go to Declaration"

Line 2 Column 16 Insert Modified Windows